

(12) **United States Patent**
Yasukawa et al.

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(54) **CONTROL APPARATUS FOR GATEWAY IN MOBILE COMMUNICATION SYSTEM**

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(71) Applicant: **SORACOM, INC.**, Tokyo (JP)

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(63) Continuation of application No. 15/764,122, filed on Mar. 28, 2018, now Pat. No. 11,805,429, which is a
(Continued)

(57) **ABSTRACT**

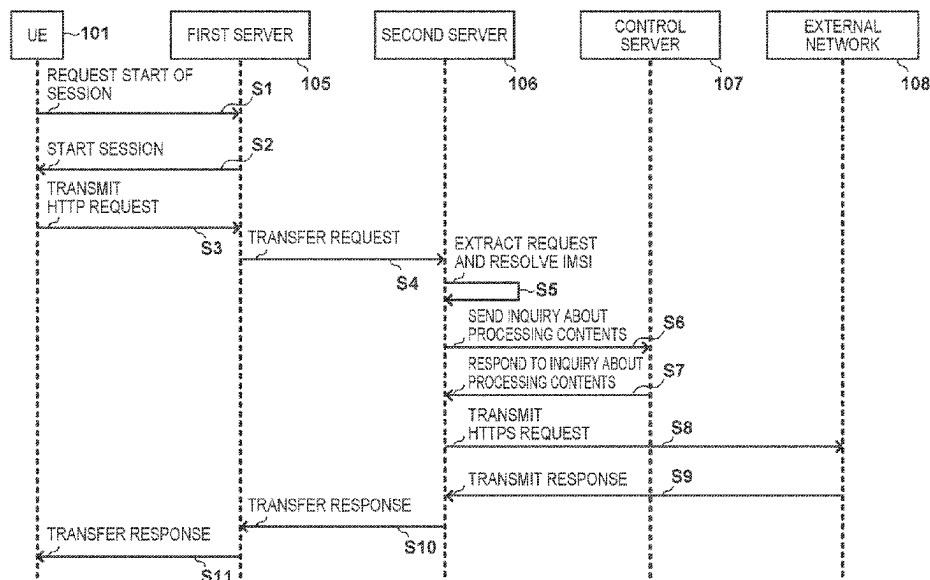
A control apparatus for a gateway functioning as the end-point of a core network in a mobile communication system is provided. The gateway includes a first server configured to receive data from a user apparatus and one or more second servers configured to transfer the data received by the first server to an external network. The control apparatus performs an operation of changing performance of the one or more second servers, an operation of monitoring a processing status of the one or more second servers, and an operation of selecting, from the one or more second servers in accordance with the processing status of the one or more second servers, a second server as a transfer destination of the data received from the user apparatus by the first server.

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(Continued)

(58) **Field of Classification Search**
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16 Claims, 7 Drawing Sheets



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H04W 24/02 (2009.01)
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H04L 63/1483 (2013.01); **H04W 12/06**
(2013.01); **H04W 12/12** (2013.01); **H04W**
88/18 (2013.01)

(58) Field of Classification Search

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28/0268; H04W 28/16; H04W 28/10;
H04W 28/065; H04W 84/18; H04W
84/12; H04W 4/206; H04W 4/02; H04W
4/20; H04W 4/21; H04W 4/025; H04W
4/18; H04W 4/70; H04W 64/00; H04W
40/04; H04W 40/24; H04W 72/121;
H04W 72/1236; H04W 72/1273; H04L
43/0882; H04L 43/16; H04L 67/1031;
H04L 67/2823; H04L 67/1006; H04L
67/18; H04L 67/2804; H04L 67/1008;
H04L 69/08; H04L 12/66; H04L 63/0272;
H04L 63/083; H04L 63/1483; H04L
47/125; H04L 47/2441; H04L 47/726;
H04L 45/38; H04L 45/58; H04L 45/302;
H04L 45/00; H04L 27/06; H04L 41/0695;
H04L 29/08; H04L 29/12; H04L 5/00;
H04B 7/0885; H04B 7/022; H04B 1/0003
See application file for complete search history.

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FIG. 1

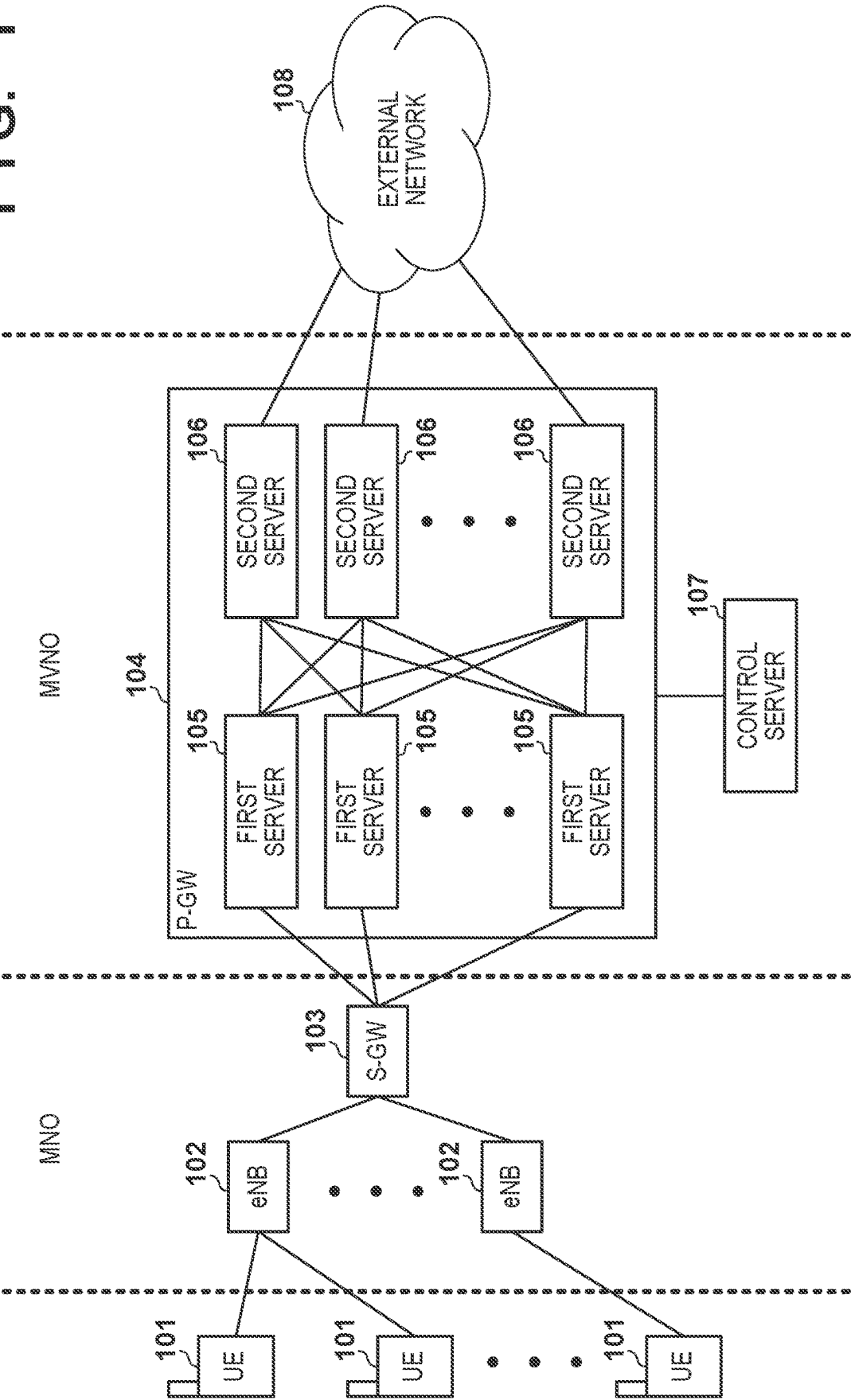


FIG. 2

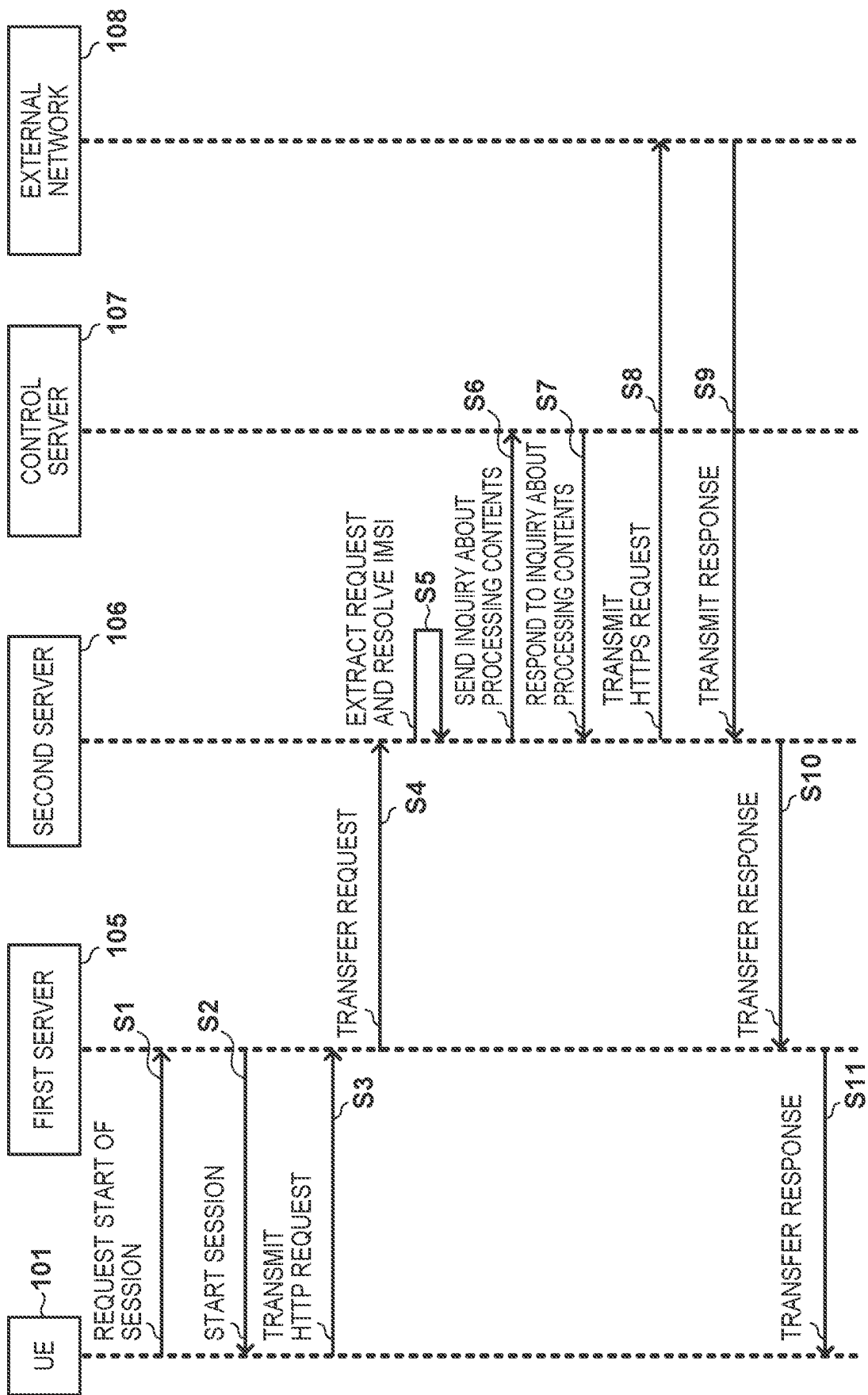


FIG. 3

300

301	302	303	304
IMSI	PROTOCOL BEFORE CONVERSION	PROTOCOL AFTER CONVERSION	DESTINATION
IMSI_A	HTTP	HTTPS	https://server1.example.com/
IMSI_B	HTTP	HTTPS	https://server2.example.com/
IMSI_C	HTTP	HTTP	http://server3.example.com/
IMSI_D	mqtt	mqtt	***
IMSI_E	TCP	TCPS	***
IMSI_F	Raw TCP socket	HTTPS	***
***	***	***	***

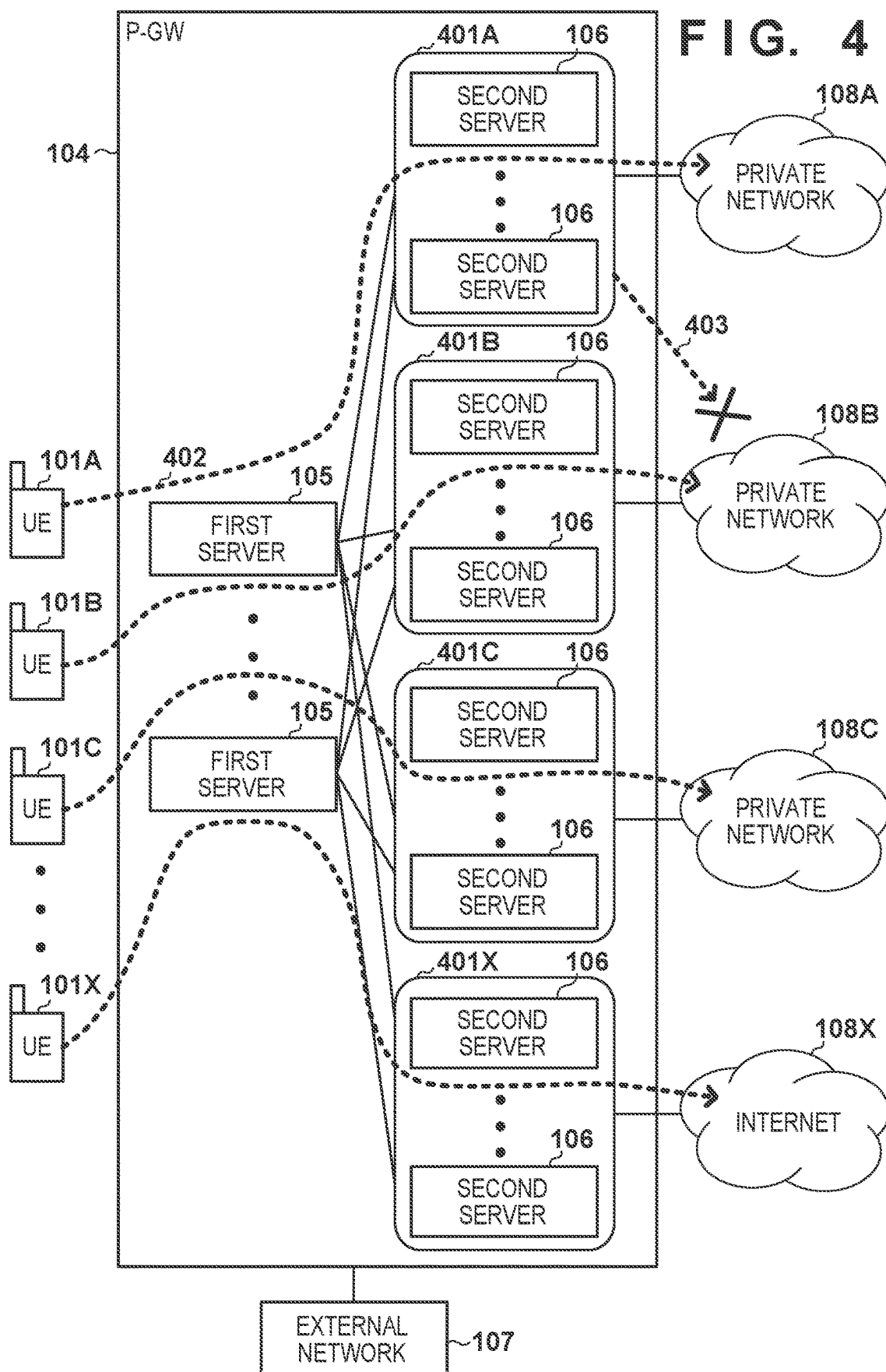


FIG. 5A

500

501 IMSI	502 GROUP ID
IMSI_1A	Group A
IMSI_1B	Group B
IMSI_1C	Group C
IMSI_4X	Default
IMSI_4Y	Default
IMSI_4Z	Default
...	...

FIG. 5B

510

511 NODE ID	512 TRANSPORT IP ADDRESS	513 USER IP ADDRESS	514 GROUP ID	515 SUBNET	516 TRANSFER DESTINATION
Node 1	169.254.240.1	172.24.240.100	Default	default_1	Internet Only
Node 2	169.254.241.1	172.24.240.100	Default	default_2	Internet Only
Node 3	169.254.240.2	172.24.240.100	Default	default_1	Internet Only
Node 4	169.254.241.2	172.24.240.100	Default	default_2	Internet Only
Node A1	169.254.242.1	172.24.240.100	Group A	subnet A1	VPC Peering A
Node A2	169.254.243.1	172.24.240.100	Group A	subnet A2	VPC Peering A
Node B1	169.254.244.1	172.24.240.100	Group B	subnet B1	Direct Connect B
Node B2	169.254.245.1	172.24.240.100	Group B	subnet B2	Direct Connect B
Node C1	169.254.246.1	172.24.240.100	Group C	subnet C1	VPN connection C
Node C2	169.254.247.1	172.24.240.100	Group C	subnet C2	VPN connection C
...	...	...	...	...	...

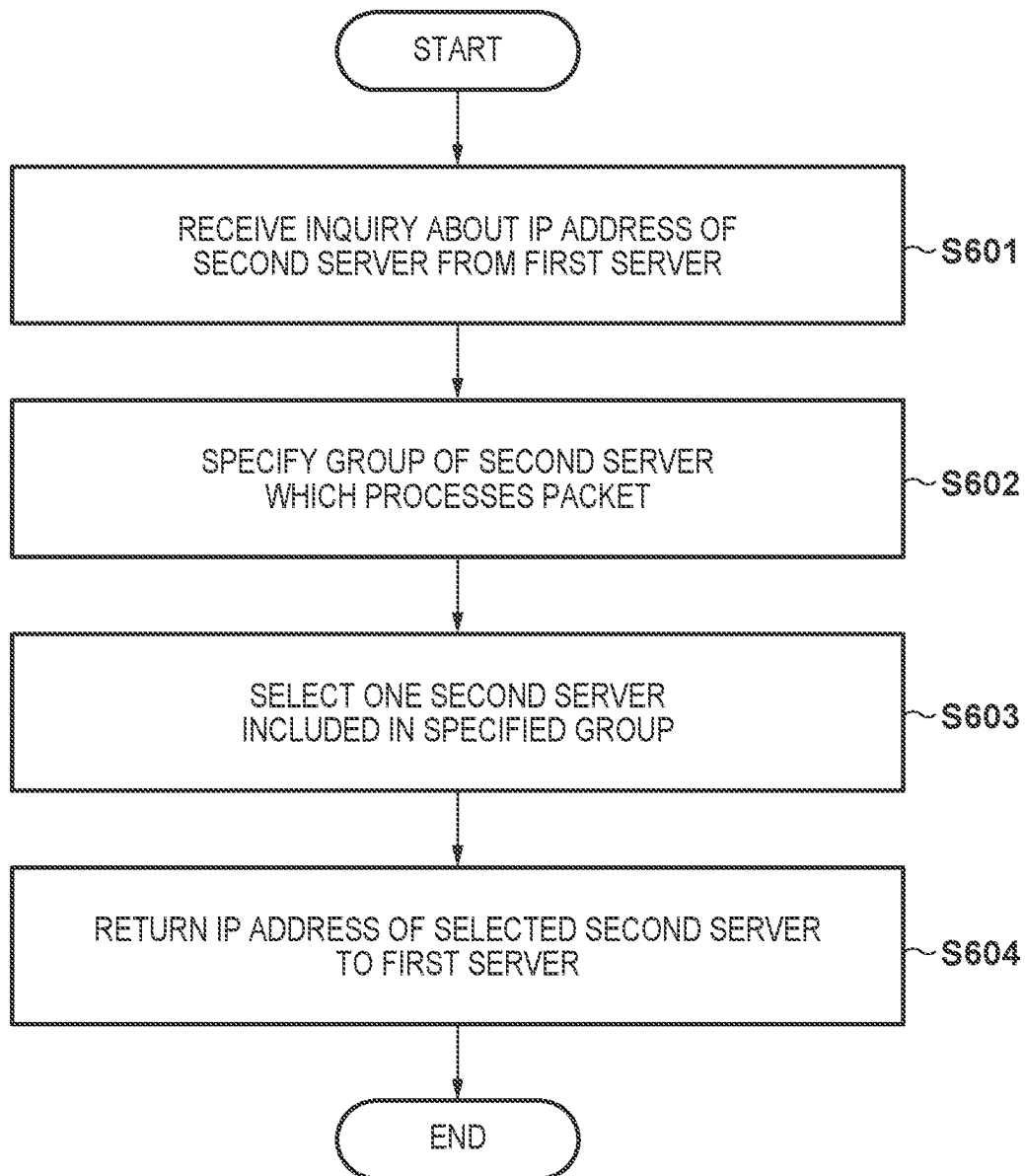
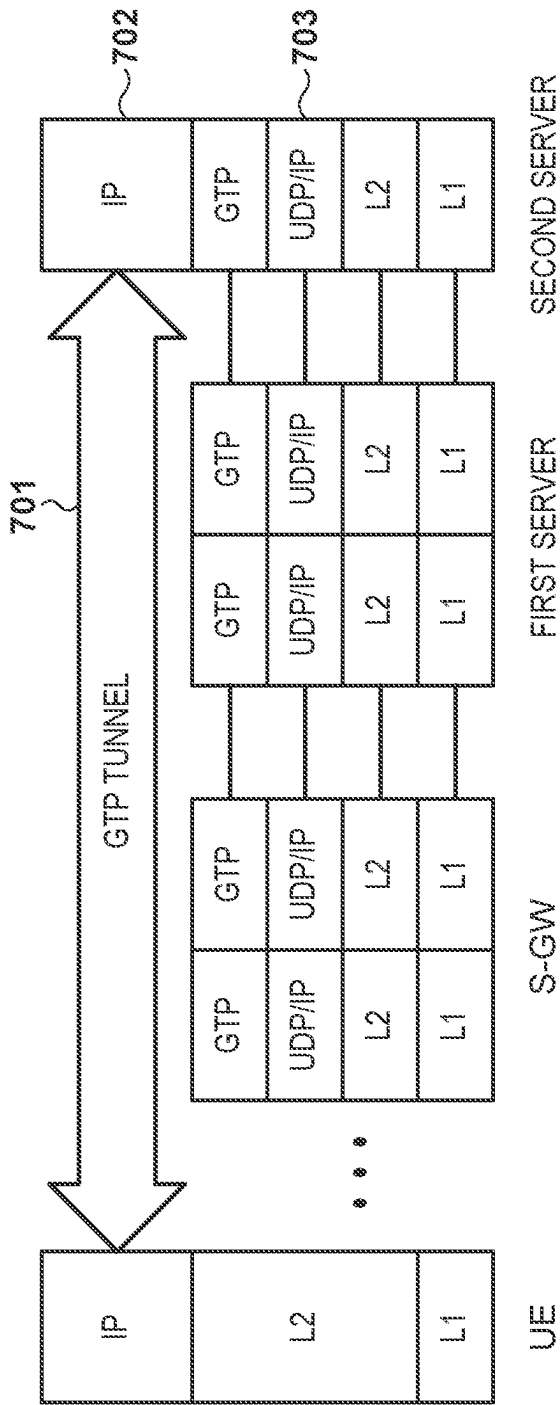
FIG. 6

FIG. 7



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CONTROL APPARATUS FOR GATEWAY IN MOBILE COMMUNICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation of U.S. application Ser. No. 15/764,122, filed Mar. 28, 2018, which claims priority to, and the benefit of, International Patent Application No. PCT/JP2015/077610 filed on Sep. 29, 2015, where the entire disclosure of both of the above applications are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a control apparatus for a gateway in a mobile communication system.

BACKGROUND ART

In recent years, mobile communication services provided by an MVNO (Mobile Virtual Network Operator) have been widespread. The MVNO provides access to an external network to a UE (User Equipment) using network components owned by an MNO (Mobile Network Operator), and a gateway apparatus of MVNO. In addition, cloud services such as an AWS (Amazon Web Service) have also been spread. The AWS can change the number of servers to be used and their performance on demand and can provision a private network (closed network) in the cloud.

SUMMARY OF INVENTION

It is assumed that a gateway apparatus which provides an MVNO is created on a cloud using cloud characteristics, and the processing capability of the gateway apparatus is changed in accordance with a communication demand. If an existing gateway apparatus is simply provided on the cloud, it is difficult to provide a service based on a demand from users. An aspect of the present invention provides an object of solving various problems when providing a gateway apparatus in a mobile communication system on a cloud.

According to some embodiments of the present invention, there is provided a control apparatus for a gateway functioning as the endpoint of a core network in a mobile communication system, wherein the gateway includes a first server configured to receive data from a user apparatus and one or more second servers configured to transfer the data received by the first server to an external network, and the control apparatus performs an operation of changing performance of the one or more second servers, an operation of monitoring a processing status of the one or more second servers, and an operation of selecting, from the one or more second servers in accordance with the processing status of the one or more second servers, a second server as a transfer destination of the data received from the user apparatus by the first server.

Other features and advantages of the present invention will be apparent from the following descriptions taken in conjunction with the accompanying drawings, in which like reference characters designate the same or similar parts throughout the figures thereof.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodi-

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ments of the invention and, together with the description, serve to explain the principles of the invention.

FIG. 1 is a block diagram for explaining an example of the arrangement of a mobile communication system according to some embodiments;

FIG. 2 is a timing chart for explaining an example of the operation of the mobile communication system according to some embodiments;

FIG. 3 explains an example of a management table of the mobile communication system according to some embodiments;

FIG. 4 is a block diagram for explaining an example of the arrangement of a mobile communication system according to some embodiments;

FIG. 5A explains an example of a management table in the mobile communication system according to some embodiments;

FIG. 5B explains an example of a management table in the mobile communication system according to some embodiments;

FIG. 6 is a flowchart for explaining an example of the operation of a control server according to some embodiments; and

FIG. 7 is a block diagram for explaining an example of the protocol configuration of the mobile communication system according to some embodiments.

DESCRIPTION OF EMBODIMENTS

The arrangement of a mobile communication system according to some embodiments of the present invention will be described with reference to FIG. 1. Although a mobile communication system complying with LTE will be described with reference to FIG. 1, the present invention is also applicable to a mobile communication system complying with 3G or any other standard. The mobile communication system mainly includes eNBs 102, an S-GW 103, a P-GW 104, and a control server 107. The eNBs 102 and the S-GW 103 are provided by an MNO (Mobile Network Operator), and the P-GW 104 and the control server 107 are provided by an MVNO (Mobile Virtual Network Operator). The P-GW 104 and the control server 107 may be created by the MVNO itself, or may be created and operated by an MVNE (Mobile Virtual Network Enabler) according to an MVNO request. The S-GW 103 may be provided by the MVNO in place of the MNO.

UEs 101 are devices to which IMSIs (International Mobile Subscriber Identities) are assigned and can be, for example, mobile phones, personal computers, sensors, and actuators. The IMSI is identification information uniquely assigned to each device and is incorporated in the device in a state in which the IMSI is stored in, for example, a SIM.

Each eNB 102 connects each UE 101 to the S-GW 103 and transfers a packet between the UE 101 and the S-GW 103. Each eNB 102 further provides a wireless resource management function, a mobility management function, and a scheduling function to each UE 101. The S-GW 103 provides a function of routing a packet from or to each UE 101. The S-GW 103 corresponds to the SGSN (packet access control node) of a 3G network.

The P-GW 104 has a function of providing access to an external network 108 to each UE 101. The P-GW 104 is a gateway apparatus functioning as the endpoint of a core network included in the mobile communication system. The external network 108 may be a public network such as the Internet or a private network provided by each individual enterprise. The P-GW 104 corresponds to the GGSN (packet

gateway node) of the 3G network. A packet from each eNB 102 to the P-GW 104 is transferred in an encapsulated form via a tunnel (GTP tunnel) established in accordance with a GTP (GPRS Tunneling Protocol). Any other L2 tunnel may be used in place of the GTP tunnel.

The P-GW 104 is formed from one or more first servers 105 and one or more second servers 106. Each first server 105 is connected to the S-GW 103 and transfers a packet received from the S-GW 103 to any one of the one or more second servers 106. Each first server 105 transfers, to the S-GW 103, a packet received by one of the one or more second servers 106.

Each second server 106 provides various services to the communications by the UE 101. For example, the second server 106 functions as NAT which performs processing in a network layer or a proxy which performs processing in an application layer. In addition, the second server 106 may perform processing for a still image and a moving image or credential assignment processing in place of the UE 101.

The upper limit of the number of servers simultaneously connectable to the S-GW 103 in the P-GW 104 is determined by the MNO. To solve this problem, the P-GW 104 is arranged as follows. The servers are classified into servers (first servers 105) which perform packet (data) exchange with the S-GW 103 and servers (second servers 106) which perform access to the external network 108 and provide various additional services, thereby increasing the number of second servers 106 so as to exceed the upper limit of connectable servers by the MNO.

Each of the one or more first servers 105 and each of the one or more second servers 106 can be virtual machines, respectively, on a cloud. This cloud may be a public cloud such as an AWS or a private cloud provided for one enterprise. If the cloud is a public cloud, the one or more first servers 105 and the one or more second servers 106 may belong to a virtual private network on the cloud. For example, if the cloud is an AWS, a virtual private network is provided by a VPC (Virtual Private Cloud) function.

The provision of the P-GW 104 on the cloud can change the performance of the P-GW 104 at an appropriate timing in accordance with the processing status of the P-GW 104. The change in performance of the P-GW 104 can be implemented by replacing (so-called scale up/scale down) one server with another server (a server having processing capability higher or lower than the previous server) or implemented by changing (so-called scale out/scale in) the number of servers.

The control server 107 is a server which controls the operation of the P-GW 104. The control server 107 may be a virtual machine on a cloud or an on-premise server. The control server 107 may be formed from a single server or a plurality of servers. In this embodiment, the control server 107 is exemplified as a server different from the first server 105 and the second server 106. However, one of the first servers 105 or one of the second servers 106 may have a function of the control server 107.

The control server 107 monitors the processing status of each of the one or more first servers 105 and each of the one or more second servers 106. Examples of the processing status of a monitoring target can be a CPU use rate, a memory use rate, a network use rate, an average throughput, the number of GTP sessions, and the like. A criterion of the processing status used to change the performance of the first server 105 may be different from a criterion of the processing status used to change the performance of the second server 106. Since the first server 105 performs the function of transferring a packet to the second server 106, the

network performance tends to become a bottleneck. For this reason, the control server 107 may change the performance of the first server 105 using the network performance (for example, the network use rate, the average throughput, or the like) as a criterion. Since the second server 106 performs various processing operations for a packet, and its calculation performance tends to become a bottleneck. For this reason, the control server 107 may change the performance of the first server 105 using the calculation function (for example, the CPU use rate, the memory use rate, or the like) as a criterion.

The control server 107 may change the performance of the P-GW 104 regardless of the processing status of the P-GW 104. For example, the control server 107 may change the performance of the P-GW 104 in accordance with an instruction from the administrator or user of the mobile communication system. In place of this, the control server 107 may automatically change the performance of the P-GW 104 in accordance with the prescheduled settings.

Subsequently, an example of a protocol configuration of a U plane (user plane) of the mobile communication system in FIG. 1 will be described with reference to FIG. 7. An end-to-end GTP tunnel 701 is established between the UE 101 and the second server 106. A packet transmitted from the UE 101 is transferred to the second server 106 via the GTP tunnel 701. An IP address (to be referred to as a user IP address hereinafter) used in an IP connection 702 with the UE 101 and an IP address (to be referred to as a transport IP address hereinafter) used in an IP connection 703 with the first server 105 are assigned to the second server 106. An IP packet is transferred from the UE 101 via the IP connection 702, and a GTP packet is transferred from the first server 105 via the IP connection 703.

Since the virtual machines in the virtual private network communicate with each other depending on a cloud service, an IP address in a subnetwork assigned to the virtual private network must be set in each virtual machine. The upper limit of the number of IP addresses in the subnetwork is determined by a cloud service. If this IP address in the subnetwork is assigned to the UE 101, the number of UEs 101 connectable to the mobile communication system is limited by the number of IP addresses in the virtual private network. For this reason, an address irrelevant to the address in the virtual environment subnet is assigned to the UE 101, NAT is performed at the virtual environment address assigned to a node when IP communication is performed with the outside of the second server 106 which performs GTP tunnel termination. In addition, in the second server 106, an IP address in the subnetwork assigned to the virtual private network is set as a transport IP address for causing the first server 105 to access the second server 106. An IP address not limited by the virtual private network is set as a user IP address for causing the UE 101 to access the second server 106. If the user IP address is additionally assigned to the second server 106 which performs GTP tunnel termination, the communication from the UE 101 to the user IP address of the second server 106 can reach the second server 106 without any NAT.

Subsequently, an example of the operation of the mobile communication system in FIG. 1 will be described with reference to FIG. 2. This operation can be performed by causing a processor such as a CPU of each UE or each server to execute a program. In place of this, each UE and each server may include circuits (for example, ASICs) for executing the following operation.

In S1, the UE 101 is attached to the eNB 102 and requests to generate a session with the P-GW 104 to the core network.

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The UE **101** includes its own IMSI (to be referred to as an IMSI_A hereinafter) in this request. The S-GW **103** establishes a C-plane GTP tunnel (GTP-C) with one of the one or more first servers **105** and transfers the request from the UE **101** to the first server **105** via this GTP tunnel. For example, the S-GW **103** selects, by a round robin method, one first server **105** from the one or more first servers **105** connected to the S-GW **103**.

Upon reception of the request from the UE **101**, the first server **105** determines whether the first server **105** refers to customer information and a session is established for the IMSI included in the request. This determination may be performed by causing the first server **105** to refer to the customer information or performed by the control server **107** which has received the request from the first server **105**. If the session can be established, the first server **105** assigns an IP address (to be referred to as 172.24.240.100) to the UE **101** and returns this IP address to the UE **101** in S2.

In S2, the first server **105** sends, to the control server **107**, a request for assignment of the second server **106**. In response to this request, the control server **107** selects one second server **106** of the one or more second servers **106** and returns the transport IP address of this second server **106** to the first server **105**. The control server **107** may select one second server **106** (for example, a server having a minimum load) in accordance with the processing status of the one or more second servers **106**. The first server **105** establishes a U-plane GTP tunnel (GTP-U) between the UE **101** and the selected second server **106**. In addition, the first server **105** notifies each second server **106** of information concerning a pair of the IMSI of the UE **101** and the IP address assigned to the UE **101**. This notification can be directly performed from the first server **105** to the second server **106** or may be performed via the control server **107**. For example, the first server **105** may register information concerning the pair of the IMSI and the IP address in the control server **107**, and the second server **106** may read out this information from the control server **107**.

In S3, the UE **101** transmits, to the first server **105** via the GTP tunnel, an HTTP request toward the user IP address (to be referred to as 169.254.254.169) of the second server **106**. The UE **101** sets the IP address (172.24.240.100) assigned in S2 in the transmission source IP address of this HTTP request. The user IP address of the second server **106** may be stored in the UE **101** before the start of the operation in S1 or may be notified from the first server **105** to the UE **101** in S2. A single IP address may be shared as the user IP addresses of the plurality of second servers **106**.

In S4, the first server **105** transfers, to the second server **106** via the GTP tunnel established in S2, the GTP packet transmitted from the UE **101** via the GTP tunnel.

In S5, the second server **106** terminates the GTP tunnel to extract the IP packet from the GTP packet. In addition, the second server **106** resolves the IMSI of the UE **101** which has transmitted the HTTP request by using the information concerning the pair notified in S2 and the transmission source IP address of this HTTP request. In place of this, the second server **106** may transmit this transmission source IP address (the IP address of the UE **101**) to the control server **107**, cause the control server **107** to resolve the IMSI, and return it to the second server **106**.

In S6, the second server **106** sends an inquiry about processing to be executed to the resolved IMSI to the control server **107**. This processing includes processing to data transmitted from the UE **101** to the external network **108** and processing to data returned from the external network **108** to the UE **101**. The control server **107** manages a table **300**

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shown in FIG. 3. Each entry of the table **300** indicates the processing contents to be executed for the data from the UE **101**. A column **301** indicates the IMSI of the UE **101** as the processing target. A column **302** indicates a request protocol before conversion. A column **303** indicates a request protocol after conversion. A column **304** indicates a request destination. For example, the first-row entry of the table **300** indicates that an HTTP request from the UE **101** having the IMSI_A is converted into an HTTPS request and the converted request is transferred to <https://server1.example.com/>. The second server **106** can perform individual processing for each UE **101**. For example, the second server **106** can similarly perform conversion from mqtt to mqtt, conversion from TCP to TCPS, and conversion from WebSocket to WebSocket over SSL. In addition to simple encryption, the second server **106** can perform conversion of a protocol itself such as HTTPS transfer of data received by a Raw TCP socket or data received by a UDP packet, and communication optimization between the UE **101** and the second server **106** such as compression of response contents from the external server or deletion of unnecessary header information. The latter function is particularly advantageous in a device such as a sensor or actuator aiming at energy saving.

In addition to the protocol conversion, the second server **106** can add authentication information such as the user name and password of a site to be accessed by the UE **101**, and add information referred to in processing on the server side, such as an identification number such as the IMEI (International Mobile Equipment Identifier) of the UE **101** and a time stamp indicating data reception time. The second server **106** can also add a digital signature using secret information exchanged in advance with the reception side so as to allow the reception side to verify the authentication of the added information. Accordingly, access by spoofing of the third party can be prevented.

The control server **107** provides an API for editing the table **300**. Via this API (Application Programming Interface), the administrator of the UE **101** can edit (for example, add, change, or delete) the operations to be performed for his own UE **101**. Similarly, by using the API, the administrator of the UE **101** can also set information such as a user name and password for authentication and secret information for verifying authenticity of the information added by the second server **106**. The control server **107** may provide a GUI (Graphical User Interface) for allowing the administrator of the UE **101** to facilitate the above processing.

In S7, the control server **107** responds, to the second server **106**, the operations to be performed of an entry indicating the result obtained by searching the table **300** using, as a key, the IMSI received from the second server **106**.

In S8, the second server **106** converts the HTTP request from the UE **101** into the HTTPS request in accordance with the operations to be performed received from the control server **107** and transfers the HTTPS request to the destination (<https://server1.example.com/>) designated by the processing contents. In S9, the second server **106** receives a response to the HTTPS request from the destination. In S10, the second server **106** adds a GTP header to this response and transfers the resultant data to the first server **105**. The first server **105** as the transfer destination is the same as the first server **105** from which the HTTP request has been transferred in S4. In S11, the first server **105** transfers the response to the UE **101** via the GTP tunnel. After that, the first server **105** may cancel the GTP tunnel between the UE **101** and the second server **106**.

According to the above processing, the common user IP address is assigned to the plurality of second servers **106**. Since the communication between the first server **105** and the second server **106** is identified by the transport IP address assigned to each second server **106**, identifiable GTP tunnels are established between the plurality of UEs **101** and the plurality of second servers **106**. For this reason, even if the scale in/scale out operation or the scale down/scale up operation of the second server **106** is performed, the endpoint used as the destination of the UE **101**, that is, the user IP address of the second server **106** need not be changed.

According to the above processing, since the user IP address of the second server **106** can be set in advance in the UE **101**, the overhead concerning the DNS processing for obtaining this IP address can be suppressed. In addition, the user IP address of the second server **106** can be set as a hardcode in the UE **101**. This is particularly advantageous when the UE **101** is a weak embedded device such as a sensor. Even if the user IP address of the second server **106** set in the UE **101** cannot be changed, the communication partner of the UE **101** in the external network **108** can be changed by updating the table **300** of the control server **107**.

Subsequently, communication between the first server **105** and the second server **106** in the processing of FIG. **2** will be described in detail. Although only one S-GW **103** is illustrated in FIG. **1**, a general mobile communication system includes a plurality of S-GWs **103**. A GTP tunnel between the S-GW **103** and the P-GW **104** (the first server **105** in practice) can be identified by a TEID (Tunnel Endpoint Identifier). Each first server **105** uniquely sets the TEID of the GTP tunnel with the S-GW **103** to prevent the interference between the GTP tunnels. However, if the respective first servers **105** set the same TEID for the respective GTP tunnels and packets are transferred to the same second server from the respective GTP tunnels, the interference occurs between the GTP tunnels.

To solve this problem, the control server **107** may monitor the TEID assigned by each first server **105** and may select the second server **106** to which the packet is transferred from the first server **105** so as to prevent the interference between the GTP tunnels in the second servers **106**. More specifically, the control server **107** may select the second server **106** to which the packet is transferred from the first server **105** so as to prevent the TEIDs assigned by the respective servers **105** from overlapping in the second servers **106**.

In place of this, another tunnel different from the GTP tunnel between the S-GW **103** and the first server **105** may be set between the first server **105** and the second server **106**, and a packet may be transferred from the first server **105** to the second server **106** via the other tunnel. The tunnel between the first server **105** and the second server **106** may be a GTP tunnel. Accordingly, the same software stack can be used in the first server **105** and the second server **106**.

In the above embodiment, the second server **106** terminates the GTP packet from the UE **101** to extract the IP address included in this GTP packet. In place of this, a third server different from the second server **106** may terminate a GTP packet from the UE **101**. In this case, the UE **101** sets the destination IP address of the IP packet as the user IP address of the third server. The second server **106** transfers, to the third server via the GTP tunnel or the other tunnel, the GTP packet transferred from the first server **105**.

Subsequently, the arrangement of a mobile communication system according to another embodiment of the present invention will be described with reference to FIG. **4**. The mobile communication system in FIG. **4** is different from the

mobile communication system in FIG. **1** in the arrangement of a P-GW **104** and the operation of a control server **107**. The rest may be the same as in the mobile communication system in FIG. **1**. In FIG. **4**, some constituent elements which are the same as those of the mobile communication system in FIG. **1** will not be illustrated. The same description as in FIG. **1** will be omitted.

A user having a UE **101** connectable to the mobile communication system may want to access his own private network using this UE **101** without going through the public network such as the Internet. For example, assume that a given enterprise purchases a plurality of SIMs and UEs assigned with these SIMs access the private network of this enterprise. Even if a second server **106** belongs to the virtual private network on a cloud, the second server **106** can access another private network by a function provided by the cloud. The accessible other private network is another virtual private network present on the same cloud as the second server **106** and different from the virtual private network to which the second server **106** belongs, or a private network of the on-premise or another cloud connected by a leased line to the virtual private network to which the second server **106** belongs.

The UE **101** of another user having no access right to a given private network of a given user should not access this private network. Due to the limitation of the cloud service, only single transfer information may be set for all servers belonging to the same subnetwork in the virtual private network. The transfer information here is information for defining an external network which the UE **101** can access. The transfer information includes, for example, a routing policy defined by a routing table, and an NACL (Network Access Control List) for access limitation. In the embodiment shown in FIG. **1**, if all the second servers **106** are included in the same subnetwork, the single transfer information is applied to all the UEs **101** which can use this mobile communication system. As a result, the UEs **101** which can access the external private network cannot be limited.

In this embodiment, the plurality of second servers **106** are divided into a plurality of groups, the respective groups are included in different subnetworks, and the transfer information is defined for each group. In the embodiment shown in FIG. **4**, the plurality of second servers **106** are classified into groups **401A**, **401B**, **401C**, and **401X**. Each group includes at least one second server **106**. As in the embodiment shown in FIG. **1**, the control server **107** can change the performance of each second server **106** included in each group, as needed. The second servers **106** belonging to the group **401A** can access only a private network **108A**. For example, the private network **108A** is another virtual private network of the same cloud as the second servers **106**. If this cloud is the AWS, these virtual private networks can be implemented by the VPC function and are connected by the VPC peering function. The second servers **106** included in the group **401B** can access only a private network **108B**. For example, the private network **108B** is an on-premise private network. If the second servers **106** are included in the AWS, the second servers **106** and the private network **108B** are connected by dedicated lines by the AWS Direct Connect function. The second servers **106** included in the group **401C** can access only a private network **108C**. For example, the private network **108A** is another virtual private network of the same cloud as the second servers **106**. If this cloud is the AWS, these virtual private networks are implemented by the VPC function and are connected by the VPN connection function. The private networks **108A** to **108C** are networks

managed by different users. The second servers **106** included in the group **401X** can access only the Internet **108X**. The one or more first servers **105** may belong to a subnetwork different from those of the second servers **106** or may belong to a virtual network different from those of the second servers **106**.

Subsequently, the above transfer information will be described with reference to FIGS. **5A** and **5B**. The transfer information is formed from two tables **500** and **510** and managed by the control server **107**. The table **500** defines that a packet from each UE **101** should be processed by the second server **106** of a specific group. Each entry (row) of the table **500** is generated for each IMSI. A column **501** indicates an IMSI. A column **502** indicates the group of the second server **106** which processes a packet from the UE **101** having each IMSI.

The table **510** indicates information concerning second servers **106**. Each entry (row) of the table **510** is generated for each second server **106**. A column **511** indicates an identifier of the second server **106**. The identifier of the second server **106** is unique in the mobile communication system. A column **512** indicates the transport IP address of the second server **106**. The transport IP address is set for each second server **106**. A column **513** indicates the user IP address of the second server **106**. The common user IP address may be set for the plurality of second servers **106** or user IP addresses may be set for the plurality of second servers, respectively, as in the above embodiment. If the user IP address of the second server **106** is set in the UE **101** in advance, the table **510** need not include the column **513**. A column **514** indicates the identifier of a group to which the second server **106** belongs. The group identifier is unique in the mobile communication system. A column **515** indicates a subnetwork to which the second server **106** belongs. A column **516** indicates the external network of packet transfer destination of the second server **106**. The common transfer destination is set for the second servers **106** belonging to the same group. The administrator (for example, MVNO) of the mobile communication system creates and updates the tables **500** and **510** in accordance with a request from the user of the mobile communication system.

Subsequently, an example of the operation of the control server **107** will be described with reference to FIG. **6**. This operation can be done by causing a processor such as the CPU of the control server **107** to execute a program. In place of this, the control server **107** may include a circuit (for example, ASIC) which execute the following operation. The operation in FIG. **6** is performed in **S2** of FIG. **2**. In the following description, a case in which the first server **105** sends, to the control server **107**, an inquiry about the transport IP address of the second server **106** and the user IP address of the second server **106** will be described below. As described above, however, the user IP address of the second server **106** may be set in the UE **101** in advance. In this case, processing about the user IP address of the second server **106** in the following description may be omitted.

In **S601**, the control server **107** receives an inquiry to obtain the transport IP address and user IP address of the second server **106** from the first server **105**. This inquiry includes the IMSI of the UE **101** received from the UE **101** in **S1** of FIG. **2** to the first server **105**.

In **S602**, the control server **107** specifies the group of the second servers **106** which are to process a packet from the UE **101** having the IMSI received in **S601** with reference to the table **500**. For example, if the IMSI received by the control server **107** is "IMSI_1A", the control server **107**

specifies that the second servers **106** included in "Group A" should process the packet from the UE **101** having this IMSI.

In **S603**, the control server **107** specifies the second servers included in the specified group and selects one of the second servers with reference to the table **510**. For example, if the group specified by the control server **107** is "Group A", the control server **107** specifies "Node A1" and "Node A2" as the second servers **106** included in this group. The control server **107** may select one second server **106** based on the processing status of each second server **106** included in the specified group. For example, the control server **107** selects the second server **106** having a lowest load.

In **S604**, the control server **107** reads out the transport IP address and user IP address of the selected second server **106** from the table **510** and returns this transport IP address to the first server **105** as the inquiry source.

In **S3** of FIG. **2**, the first server **105** then notifies the UE **101** of the user IP address of the selected second server **106**. In **S4** of FIG. **2**, the UE **101** transmits, to the first server **105**, the HTTP request toward the user IP address of the selected second server **106**. In **S5** of FIG. **2**, the first server **105** transfers the packet to the selected second server **106** using the transport IP address of the second server **106** obtained in **S604**. After that, the second server **106** transfers the packet to the external network **108** in accordance with the transfer information set in the subnetwork to which this second server **106** belongs.

The packet from a UE **101A** whose IMSI is "IMSI_1A" is transferred to the second servers **106** included in the group **401A**. Since the second servers **106** included in the group **401A** can access only the private network **108A**, the packet from the UE **101A** is transferred to only the private network **108A**, as indicated by a dotted line **402**, but is not transmitted to other private networks, as indicated by a dotted line **403**. Similarly, a UE **101B** is transferred to only the private network **108B** via the second servers **106** included in the group **401B**. A UE **101C** is transferred to only the private network **108C** via the second servers **106** included in the group **401C**. A UE **101X** in which access to a private network is not set is transferred to only the Internet **108X** via the second servers **106** included in the default group **401X**.

In the embodiment shown in FIG. **4**, the second server **106** terminates the GTP tunnel, extracts the IP packet, and then transfers the IP packet to the external network via the L3 connection. However, depending on a cloud service, the number of peer-connectable virtual networks and the number of virtual interfaces for private connections are limited. As a result, the number of private networks connectable to the P-GW **104** is limited. To solve this problem, the second server **106** may set the L2 tunnel with the server in the external private network and transfer the packet via the L2 tunnel. If the L2 tunnel is used, processing using the IP address of the UE **101** can be performed on the private network side. Whether the L2 tunnel or L3 connection is used can be set for each private network.

In the above description, a case in which the MVNO manages the P-GW **104** has been described. However, the present invention is also applicable when the MNO manages the P-GW **104**. A case in which P-GW **104** is L2-connected to the UE **101** has been described above. However, the present invention is also applicable when the P-GW **104** is L3-connected to the UE **101**.

In each embodiment described above, the first servers **105** may be classified into a C-plane server for performing communication with the S-GW **103** on the C plane and a P-plane server for performing communication with the

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S-GW **103** on the P plane. In addition, the C-plane servers can also be classified into a server for receiving data from the S-GW **103** via the GTP tunnel and a server for decapsulating this data and extracting the IP packet.

The present invention is not limited to the above-described embodiments, and various changes and modifications can be made within the spirit and scope of the present invention. Therefore, to apprise the public of the scope of the present invention, the following claims are made.

The invention claimed is:

1. A communication system comprising a gateway for a U-plane and a control apparatus for controlling the gateway, wherein

the gateway includes a first server for the U-plane configured to receive data from a user apparatus and two or more second servers for the U-plane configured to transfer the data received by the first server to an external network,

the control apparatus performs

an operation of monitoring a processing status of the two or more second servers, and

an operation of selecting, from the two or more second servers in accordance with the processing status of the two or more second servers, a second server as a transfer destination of the data received from the user apparatus by the first server, and

each of the two or more second servers is assigned with an IP address for use in an IP connection for the U-plane with the first server.

2. The system according to claim **1**, wherein the second server selected by the control apparatus performs processing in one of a network layer and an application layer for data transferred from the first server.

3. The system according to claim **1**, wherein the second server selected by the control apparatus transfers, to another server configured to perform processing in one of a network layer and an application layer for data, the data transferred from the first server.

4. The system according to claim **1**, wherein the first server receives data from the user apparatus via an L2 tunnel, and

the first server transfers the data, via an L2 tunnel different from the L2 tunnel, to the second server selected by the control apparatus.

5. The system according to claim **1**, wherein the gateway includes a plurality of first servers, the first server receives data from the user apparatus via an L2 tunnel, and

the control apparatus selects the second server serving as a transfer destination of data received from the user apparatus by the first server so as not to overlap identifiers of the L2 tunnels in the second server.

6. The system according to claim **1**, wherein the gateway includes two or more first servers, the two or more first servers are virtual machines on a cloud,

the control apparatus further performs

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an operation of changing performance of the two or more second servers, and

an operation of monitoring a processing status of the two or more second servers, and

the control apparatus performs a change in performance of the two or more first servers and the change in performance of the two or more second servers based on different criteria.

7. The system according to claim **6**, wherein the control apparatus performs the change in the performance of the two or more first servers based on network performance, and the change in the performance of the two or more second servers based on calculation performance.

8. The system according to claim **1**, wherein the second server and the external network communicate with each other via the L2 tunnel.

9. The system according to claim **1**, wherein the second server and the external network communicate with each other via an L3 connection.

10. The system according to claim **1**, wherein the two or more second servers are virtual machines on a cloud.

11. The system according to claim **1**, wherein the gateway is a P-GW or GGSN.

12. The system according to claim **1**, wherein the control apparatus further performs an operation of changing performance of the two or more second servers.

13. The system according to claim **1**, wherein at least two respective IP addresses are for use by individual ones of the second server,

the respective IP addresses of the two or more second servers are different from each other,

the control apparatus selects one second server from the two or more second servers, and

the first server transmits the data to the respective IP address of the selected second server.

14. The system according to claim **1**, wherein each of the two or more second servers is assigned with a first IP address for use in an IP connection with the user apparatus and a second IP address for use in an IP connection with the first server.

15. The system according to claim **14**, wherein at least two second respective IP addresses are for use by individual ones of the second servers as the second IP addresses,

the second respective IP addresses of the two or more second servers are different from each other,

the control apparatus selects one second server from the two or more second servers, and

the first server transmits the data to the second respective IP address of the selected second server.

16. The system according to claim **15**, wherein the two or more second servers share the first IP address, the second respective IP address being different from the first IP address, and

the first server receives data that the user apparatus has transmitted to the first IP address that is shared by the two or more second servers.

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